

SEQUENTIAL INFORMATION CASCADE WITH MARTINGALE FAIRNESS

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1 Introduction

This paper introduces a sequential guessing game, dubbed the "Mystery Guest" game. Consider a set of N players, defined by $[N] = \{1, \dots, N\}$. Let $i \in [N]$ denote the i^{th} player. The game proceeds sequentially from player $i = 1$ to $i = N$. On their turn, each player may ask a question to gather clues before attempting to guess the identity of a historical person. If they guess correctly, they win and the game terminates. If they guess incorrectly, they lose points. If they pass, they lose nothing, and the turn shifts to the next player.

Let M be the maximum number of cycles the game runs, and L be the maximum number of guesses allowed per player (where $L \leq M$). Passing allows the next player to ask a question, which yields more information. This increase in the probability of a correct guess is modeled as a Bayesian update.

Let Φ_{ij} denote the probability that player i guesses correctly during cycle j . Let p_{ij} be the points gained by player i in cycle j for a correct answer, and n_{ij} be the points lost for an incorrect answer. Let l_i denote the remaining guesses for player i .

The expected points for player i during cycle j is:

$$E_{ij} = \Phi_{ij}p_{ij} - (1 - \Phi_{ij})n_{ij} \quad (1)$$

We define fairness as the *Invariance of the Expectation* E_{ij} . Therefore, local Martingale fairness is satisfied if:

$$\forall i, i' \in [N], \forall j, j' \in [M], \quad E_{ij} = E_{i'j'} = C \quad (2)$$

where C is the fairness constant. Assuming for the first player $p_{11} = P_0 > 0$ and $n_{11} = 0$, we have:

$$E_{11} = \Phi_{11}P_0 = C \quad (3)$$

We enforce an additive and subtractive evolution of points, setting $p_{ij} = P_0 - x_{ij}$ (where $x_{11} = 0$) and $n_{ij} = x_{ij}$. In this constrained framework:

$$\begin{aligned} E_{ij} &= \Phi_{ij}(P_0 - x_{ij}) - (1 - \Phi_{ij})x_{ij} \\ &= \Phi_{ij}P_0 - \Phi_{ij}x_{ij} - x_{ij} + \Phi_{ij}x_{ij} \\ &= \Phi_{ij}P_0 - x_{ij} = \Phi_{11}P_0 \end{aligned} \quad (4)$$

$$\Rightarrow x_{ij} = P_0(\Phi_{ij} - \Phi_{11}) \quad (5)$$

To maximize their chances of winning, a rational player aims to answer before anyone else, yielding the highest possible score. To formally define the timing of this decision, let $\Lambda_{ij} \in \{0, 1\}$ be the probability that player i answers during cycle j . Thus, $1 - \Lambda_{ij}$ is the probability of passing. The probability of answering correctly and ending the game is $\Lambda_{ij}\Phi_{ij}$, while the probability of guessing incorrectly is $\Lambda_{ij}(1 - \Phi_{ij})$.

The probability that the game terminates exactly at turn i', j' is:

$$\prod_{i,j \in [i'-1] \times [j'-1]} \left((1 - \Lambda_{ij}) + \Lambda_{ij}(1 - \Phi_{ij}) \right) (\Lambda_{i'j'}\Phi_{i'j'}) \quad (6)$$

Assuming $j < M$, the probability that the game does not end and player i gets another turn in the subsequent cycle is the survival probability S_{ij} :

$$\begin{aligned} S_{ij} &= \prod_{k \in [N] \setminus \{i\}} P(\text{player } k \text{ either loses or passes}) \\ &= \prod_{k=i+1}^N \left((1 - \Lambda_{kj}) + \Lambda_{kj}(1 - \Phi_{kj}) \right) \prod_{k=1}^{i-1} \left((1 - \Lambda_{k(j+1)}) + \Lambda_{k(j+1)}(1 - \Phi_{k(j+1)}) \right) \end{aligned} \quad (7)$$

This defines the full-capacity survival probability, assuming all players have remaining guesses ($l_k \geq M - j + 1$). To generalize this, we introduce an indicator function:

$$\mathbb{I}_k = \begin{cases} 1 & \text{if } l_k > 0 \\ 0 & \text{if } l_k = 0 \end{cases} \quad (8)$$

Using this indicator, we define a constrained survival probability $S'_{ij} \leq S_{ij}$:

$$S'_{ij} = \prod_{k=i+1}^N (1 - \Lambda_{kj} \Phi_{kj} \mathbb{I}_k) \prod_{k=1}^{i-1} (1 - \Lambda_{k(j+1)} \Phi_{k(j+1)} \mathbb{I}_k) \quad (9)$$

This constrained probability S'_{ij} acts as the foundation for constructing the Value function.

2 Optimal Strategy under Perfect Rationality

Assuming perfect rationality and complete information, we apply backward induction to determine the Subgame Perfect Nash Equilibrium (SPNE). Let $V_i(j, l_i)$ be the maximum expected value for player i from cycle j onwards.

If player i chooses to answer ($\Lambda_{ij} = 1$), they receive expected points C if correct. If incorrect (probability $1 - \Phi_{ij}$), they lose a guess, and their future expected value depends on the game surviving to cycle $j + 1$. The expected value of answering is:

$$V_{ans} = C + (1 - \Phi_{ij}) S'_{ij} V_i(j + 1, l_i - 1) \quad (10)$$

If player i passes ($\Lambda_{ij} = 0$), they forfeit the immediate payout C , relying entirely on the future value:

$$V_{pass} = S'_{ij} V_i(j + 1, l_i) \quad (11)$$

A rational player selects the action that maximizes their expectation:

$$V_i(j, l_i) = \max(V_{ans}, V_{pass}) \quad (12)$$

- If $V_{ans} \geq V_{pass}$, guessing is optimal.
- If $V_{pass} > V_{ans}$, passing is optimal.

2.1 Backward Induction

At the terminal state where $i = N, j = M$ (assuming $l_N > 0$), player N must guess:

$$V_N(M, l_N) = \max(C, 0) = C \quad (13)$$

Knowing this, player $N - 1$ makes their choice. In fact, during the final cycle M , all players with remaining guesses will strictly prefer to guess:

$$\forall i \in [N], \quad V_i(M, l_i) = C \quad (\text{assuming } l_i > 0) \quad (14)$$

Moving back to cycle $M - 1$ for player N :

$$V_N(M - 1, l_N) = \max \left(C + (1 - \Phi_{N(M-1)}) S'_{N(M-1)} V_N(M, l_N - 1), S'_{N(M-1)} V_N(M, l_N) \right) \quad (15)$$

Because $\Lambda_{iM}^* = 1$ for all players in cycle M , player N 's survival probability at $M - 1$ is:

$$S'_{N(M-1)} = \prod_{k=1}^{N-1} (1 - \Phi_{k(M)} \mathbb{I}_k) \quad (16)$$

We evaluate this across two cases for player N 's remaining guesses:

Case I: $l_N = 1$

Player N has only one guess remaining. Because $V_N(M, 0) = 0$:

$$V_N(M-1, l_N) = \max(C + 0, S'_{N(M-1)}C) = \max(C, S'_{N(M-1)}C) \quad (17)$$

Since $S'_{N(M-1)} \leq 1$, it follows that $C \geq S'_{N(M-1)}C$. Therefore, $V_{ans} \geq V_{pass}$, and player N should guess immediately.

Case II: $l_N > 1$

Player N has multiple guesses. Using $V_N(M, l_N - 1) = C$ and $V_N(M, l_N) = C$:

$$V_N(M-1, l_N) = \max\left(C + (1 - \Phi_{N(M-1)})S'_{N(M-1)}C, S'_{N(M-1)}C\right) \quad (18)$$

The player prefers to guess if $V_{ans} \geq V_{pass}$:

$$1 + (1 - \Phi_{N(M-1)})S'_{N(M-1)} \geq S'_{N(M-1)} \quad (19)$$

If $\Phi_{N(M-1)} < 1$, isolating S' yields:

$$S'_{N(M-1)} \leq \frac{1}{\Phi_{N(M-1)}} \quad (20)$$

Because $\Phi \in [0, 1]$, the term $\frac{1}{\Phi} \geq 1$. Since a probability S' cannot exceed 1, this inequality is trivially satisfied. Therefore, in all cases, guessing immediately remains the optimal strategy.

3 The Irrationality of Waiting

We formally establish the condition under which immediately guessing is the universally optimal choice for all players:

$$S'_{ij} \leq \frac{1}{1 + \Phi_{ij}(l_{ij} - 1)} \quad (21)$$

We use induction to prove that under this bound, $\Lambda_{ij}^* = 1$ for all i, j . The base case for the terminal cycle is established: $V_{iM} = V_{iM}^{ans} = C$.

Assume the inductive hypothesis for cycle $j + 1$:

$$V_{i(j+1)} = V_{i(j+1)}^{ans} = C + (1 - \Phi_{i(j+1)})S'_{i(j+1)}V_{i(j+2)} \quad (22)$$

We assert the following upper bounds on the marginal and total value functions:

$$V_i(j, l_i) - V_i(j, l_i - 1) \leq C \quad (23)$$

$$V_i(j, l_i) \leq Cl_i \quad (24)$$

Evaluating the choice at node ij :

$$V_{ij} = \begin{cases} C + (1 - \Phi_{ij})S'_{ij}V_i(j+1, l_i - 1) & \text{if } \Lambda_{ij}^* = 1 \\ S'_{ij}V_i(j+1, l_i) & \text{if } \Lambda_{ij}^* = 0 \end{cases} \quad (25)$$

For guessing to be optimal ($V_{ij}^{ans} \geq V_{ij}^{pass}$):

$$\begin{aligned} C + (1 - \Phi_{ij})S'_{ij}V_i(j+1, l_i - 1) &\geq S'_{ij}V_i(j+1, l_i) \\ C - \Phi_{ij}S'_{ij}V_i(j+1, l_i - 1) &\geq S'_{ij}(V_i(j+1, l_i) - V_i(j+1, l_i - 1)) \end{aligned} \quad (26)$$

Substituting our defined upper bounds:

$$\begin{aligned} C - C(l_i - 1)\Phi_{ij}S'_{ij} &\geq S'_{ij}C \\ \Rightarrow S'_{ij} &\leq \frac{1}{1 + \Phi_{ij}(l_i - 1)} \end{aligned} \quad (27)$$

Thus, under this threshold, passing ($\Lambda_{ij} = 0$) is strictly suboptimal, and rational players will always guess.

4 Global vs. Local Fairness Expectation

Given the SPNE where $\Lambda_{ij}^* = 1$, players will utilize all L guesses. Let Ω_m be the cumulative survival probability of player i reaching cycle m .

The expected points for player i in a single cycle m is:

$$\mathbb{E}_m = (1 - \Omega_m)(0) + \Omega_m C = \Omega_m C \quad (28)$$

The total global expectation $\mathbb{G}(i)$ across all L cycles is:

$$\mathbb{G}(i) = \sum_{m=1}^L \Omega_m C = C \sum_{m=1}^L \left[\prod_{k=1}^{(m-1)N+i-1} (1 - \Phi_k) \right] \quad (29)$$

Comparing adjacent players i and $i + 1$, we observe:

$$\begin{aligned} \mathbb{G}(i+1) &= C \sum_{m=1}^L \left[(1 - \Phi_{(m-1)N+i}) \prod_{k=1}^{(m-1)N+i-1} (1 - \Phi_k) \right] \\ &\leq C \sum_{m=1}^L \left[\prod_{k=1}^{(m-1)N+i-1} (1 - \Phi_k) \right] = \mathbb{G}(i) \end{aligned} \quad (30)$$

Because $\Phi > 0$, the expectation strictly decreases. Thus, global fairness is impossible under a constant local payout structure, confirming a strict first-mover advantage.

5 Constant Payoffs and Changing Expectations

Relaxing the Martingale constraint, assume constant payoffs across all turns: $p_{ij} = p$ and $n_{ij} = q$. The expected value of answering becomes:

$$E_{ij} = \Phi_{ij} p + (1 - \Phi_{ij}) q \quad (31)$$

The Bellman inequality ($V^{ans} \geq V^{pass}$) becomes:

$$\begin{aligned} E_{ij} + (1 - \Phi_{ij}) S'_{ij} V_i(j+1, l_i - 1) &\geq S'_{ij} V_i(j+1, l_i) \\ \Phi_{ij} p - (1 - \Phi_{ij}) q + (1 - \Phi_{ij}) S'_{ij} V_i(j+1, l_i - 1) &\geq S'_{ij} V_i(j+1, l_i) \\ \Rightarrow S'_{ij} &\leq \frac{E_{ij}}{V_i(j+1, l_i) - (1 - \Phi_{ij}) V_i(j+1, l_i - 1)} \end{aligned} \quad (32)$$

While future probabilities are obfuscated, backward induction continues to hold under this bound, enforcing the SPNE where passing is irrational.

6 The Impossibility of Dual Fairness

We evaluate whether global fairness ($\mathbb{G}(i) - \mathbb{G}(i+1) = 0$) can coexist with local fairness ($E_{im} = C$).

$$\mathbb{G}(i) - \mathbb{G}(i+1) = \sum_{m=1}^L \Omega_{im} [E_{im} - E_{(i+1)m} (1 - \Phi_{(m-1)N+i})] \quad (33)$$

Assuming local fairness ($E_{im} = E_{(i+1)m} = C$):

$$\sum_{m=1}^L \Omega_{im} [C - C(1 - \Phi_{(m-1)N+i})] = \sum_{m=1}^L \Omega_{im} C \Phi_{(m-1)N+i} \quad (34)$$

Since $\Omega_{im}, C, \Phi > 0$, this sum is strictly positive, yielding a contradiction.

To achieve global fairness, the required local expectation must satisfy:

$$E_{im} = E_{(i+1)m} (1 - \Phi_{(m-1)N+i}) \quad (35)$$

Solving recursively for the general expectation E_{im} :

$$E_{im} = \frac{E_{11}}{\prod_{t=1}^{(m-1)N+i-1} (1 - \Phi_t)} = \frac{E_{11}}{\Omega_{im}} \quad (36)$$

Theorem 1 (Impossibility of Dual Fairness) *In a sequential game with N players, M total cycles, and L guesses per player, assuming conditions that induce a Subgame Perfect Nash Equilibrium where all players choose to guess ($\Lambda = 1$), the game cannot be simultaneously locally fair (constant expected payoff per turn) and globally fair (equal total expected payoff per player).*